

Motivation Letter: Research Assistant (RE1)

Juan David Bendek Williamson

Barcelona, Spain | +34 623 124 525 | juandavidbendeck@hotmail.com

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Recruitment Panel

Barcelona Supercomputing Center - Centro Nacional de Supercomputación

Subject: Application for Research Assistant (RE1), reference 401_26_ES_ESS_RE1

Dear Members of the Recruitment Panel,

I am applying for the Research Assistant position in the Earth System Services Group because the BIG-5 project requires the kind of interdisciplinary bridge that has defined my recent work: transforming heterogeneous image and text data into computational workflows whose inputs, model behaviour, and outputs can be examined. BIG-5's study of how nature-related values form and spread across virtual communities is especially compelling because it connects machine learning with a substantive social-ecological question rather than treating automated classification as an end in itself.

I recently completed an MSc in Modelling for Science and Engineering at the Universitat Autònoma de Barcelona, specialising in Data Science, after a BSc in Mechanical Engineering. I completed my BSc under the Hungarian Government's Stipendium Hungaricum scholarship, which covered my tuition and provided living and accommodation support. I was later selected for CSIC's JAE Intro ICU 2025 research-introduction scholarship at IIIA-CSIC, where I developed my MSc thesis. This progression has trained me to move between domain questions, data, computational models, and research software.

At Aily Labs, I took responsibility for integration seams connecting go-to-market data capabilities to SuperAgent, the company's general-purpose, customer-facing LLM reasoning agent. I developed automated analyses that produced validated CSV artifacts, aggregated summaries, and structured output contracts for downstream orchestration. My work progressed from MCP server tools to task-scoped agent skills and then into a governed semantic layer. Curated Markdown resources described the meaning, relationships, permitted use, and access constraints of approved SQL tables so agents could identify and use the correct data without relying on raw schemas. I developed and maintained capabilities used by Aily tenants including Sanofi.

When answer quality regressed or a data resource, skill, or workflow contract changed, my team was responsible for evaluating and correcting the behaviour from development through production. I traced failures across data availability, table allowlists, semantic metadata, tool selection, DAG payloads, and post-processing, then verified corrections before promotion. Unit and integration tests, representative question sets, trace inspection, continuous integration, and pre-production validation provided the release evidence. This gave me practical experience with data organisation, Python backend development, NLP-enabled systems, and the maintenance of research-like model evaluations under production constraints.

My MSc thesis at IIIA-CSIC brought text and image processing into one embodied system. I designed and validated a modular ROS 2 and ROS4HRI architecture for a NAO robot that combines dialogue, language-model planning, symbolic knowledge, deterministic execution, and feedback-driven recovery. On the visual side, I trained a custom YOLO detector for a restricted object set and integrated it with the robot camera stream. The grounding pipeline normalised detections, applied confidence and label controls, maintained bounded object identities, and converted accepted observations into provenance-bearing symbolic facts. On the language side, structured dialogue and planner outputs were evaluated against requested entities, execution traces, knowledge-base postconditions, and controlled failure cases. This work taught me to treat model output as evidence that must be compared with observable and human-defined criteria.

My earlier work provides a complementary scientific foundation. My BSc thesis built and validated a neural network for turbulent-intensity prediction from wind-profile data, requiring data preparation, model selection, validation, and interpretation against a physical process. At ABS Consulting, I later supported seismic qualification and structural assessment for UK nuclear-power assets. The work required reviewable calculations, standards-based reasoning, and technical reports in which assumptions and results remained traceable.

iTrader, my autonomous trading platform under staged development, has extended my experience with data collection, model experimentation, and scientific visualisation. Its input layer supports historical OHLCV backfilling and live WebSocket ingestion, followed by solver-specific feature engineering and validated task generation. The implemented offline stack uses PPO/GAE with FiLM-conditioned policy and value networks, decomposed rewards, reproducible run manifests, flattened JSONL and CSV metrics, and automatically generated Matplotlib diagnostics. Building it has required careful dataset boundaries, experiment comparison, failure handling, and visual communication of training behaviour.

Watson, the personal agent platform I operate for software development and research, adds practical system-administration and backend experience. Hermes provides the agent harness and tool workflow, LiteLLM provides model routing and an OpenAI-compatible proxy, llama.cpp serves a quantised Qwen model on the GPU, and ZeroTier exposes the endpoint across my machines. I maintain its launch, health, routing, recovery, and performance checks. This work has made Linux, service configuration, logs, network boundaries, and reproducible model interfaces part of my normal development practice.

My Data Science specialisation also covered the computational infrastructure needed for large datasets. Parallel Programming included C, CUDA, OpenMP, MPI, SLURM, GPU programming, and performance analysis. Distributed Systems covered cluster and supercomputer infrastructure, batch workloads, distributed databases, cloud systems, and fault tolerance. Through laboratory and project work at UAB and IIIA-CSIC, I submitted and monitored cluster jobs, developed parallel solutions, and assessed computational behaviour. I would bring this foundation to BIG-5's high-performance data-processing requirements while continuing to deepen it within BSC.

I could contribute immediately to Python data preparation, backend services, NLP and image-model experimentation, system administration, and research visualisation. My formal training is computational rather than in social science, so I would approach the project's conceptual and qualitative methods in close collaboration with its domain researchers. I am prepared to extend my current software experience into the project's frontend requirements and to learn the appropriate computational social-science and social-network-analysis methods without treating them as interchangeable with general machine learning.

Within BIG-5, I would be particularly interested in building provenance-preserving datasets that connect image and text content to platform, language, landscape, human coding, model version, and uncertainty. Comparing automated assessments with human judgements, documenting sampling and annotation decisions, and producing interpretable visualisations would allow the computational work to remain accountable to the project's social-ecological questions. The project's multilingual and cross-platform scope also makes careful data governance, bias analysis, and reproducibility essential.

I work independently, document decisions carefully, and communicate directly across technical and domain teams. I am fluent in English and Spanish, live in Barcelona, and have studied or worked in Hungary, the United Kingdom, and Spain. I would welcome the opportunity to support Dr. Johannes Langemeyer's team and contribute to the Earth System Services Group's interdisciplinary research.

Thank you for considering my application. I would be pleased to discuss how my experience in data pipelines, language and vision systems, scientific machine learning, and dependable research software could contribute to BIG-5.

Yours sincerely,

Juan David Bendek Williamson

References

1. **Francisco Martin**, Principal Data Scientist, Aily Labs

Manager | francisco.martin@ailylabs.com

2. **Dr. Balogh Miklos**, Professor, BME

BSc Thesis Supervisor | balogh.miklos@gpk.bme.hu

3. **Ben Wolfe**, Senior Engineer, ABS Consulting UK

Supervisor | bwolfe@eagle.org